# MTH401 Assignment 01 Spring 2026

**Deadline: 12 May,2026**

**Total marks= 20**

**Name:**

**Student id:**

**Question # 1:** **Marks=05**

Find the orthogonal trajectories of the family of parabolas

$y = cx^2$

where $c$is an arbitrary constant.

**Answer:**

$\frac{dy}{dx} = 2cx$

$c = \frac{y}{x^2}$

$\frac{dy}{dx} = 2x\left(\frac{y}{x^2}\right)$

this is deferential equation

$\frac{dy}{dx} = \frac{2y}{x}$

replace

$\frac{dy}{dx}$ with $-\frac{1}{\frac{2y}{x}}$

*than*

$\frac{dy}{dx} = -\frac{x}{2y}$

*now*

$2y2d = -xdx$

now apply integration method

$\int 2ydy = \int -xdx$

$y^2 = -\frac{x^2}{2} + c$

now multiply by 2

$2y^2 = -x^2 + c$

$x^2 + 2y^2 = c$

*hence*

$x^2 + 2y^2 = c$

**Question # 2:** **Marks=05**

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**Answer:**

according to given statement the logistic equation is

$$\frac{dx}{dt} = kx(800-x)$$

$k$ is constant

$$\frac{dx}{x(800-x)} = kdt$$

$$\frac{1}{x(800-x)} = \frac{1}{800}\left(\frac{1}{x} + \frac{1}{800-x}\right)$$

$$\int \frac{1}{800}\left(\frac{1}{x} + \frac{1}{800-x}\right)dx = \int kdt$$

$$\frac{1}{800}(ln|x| - ln|800-x|) = kt + c$$

$$ln\left(\frac{x}{800-x}\right) = 800kt + c_1$$

$$\frac{x}{800-x} = ce^{800kt}$$

$$x = (800-x)ce^{800kt}$$

$$x + xce^{800kt} = 800ce^{800kt}$$

$$x(1+ce^{800kt}) = 800ce^{800kt}$$

$$x = \frac{800ce^{800kt}}{1+ce^{800kt}}$$

$$x = \frac{800}{1+ae^{-800kt}}$$

$$1 = \frac{800}{1+a}$$

$$1+a = 800$$

$$a = 799$$

$$x(t) = \frac{800}{1+799e^{-800kt}}$$

now we use x(3)=40 condition

$$40 = \frac{800}{1+799e^{-2400k}}$$

$$40(1+799e^{-2400k}) = 800$$

$$1+799e^{-2400k} = 20$$

$$799e^{-2400k} = 19$$

$$e^{-2400k} = \frac{19}{799}$$

now for x(7)

$$x(7) = \frac{800}{1+799e^{-5600k}}$$

$$e^{-5600k} = (e^{-2400k})^{\frac{5600}{2400}}$$

$$e^{-5600k} = \left(\frac{19}{799}\right)^{\frac{7}{3}}$$

$$x(7) = \frac{800}{1+799\left(\frac{19}{799}\right)^{\frac{7}{3}}}$$

$$\left(\frac{19}{799}\right)^{\frac{7}{3}} \approx 0.000164$$

$$799(0.000164) \approx 0.131$$

$$1+0.131 = 1.131$$

$$x(7) \approx \frac{800}{1.131}$$

*approximately*

$$x(7) \approx 707$$

**Question # 3:** **Marks=05**

Find the general solution of the differential equation.

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Identify $y_1, y_2$ and then show that both are linearly independent by Wronskian.

**Answer:**

suppose that

$y = e^{mx}$

$y'' = m^2 e^{mx}$

$m^2 e^{mx} - 9e^{mx} = 0$

$m^2 - 9 = 0$

now for m

$m^2 = 9$

$m = \pm 3$

*here*

$m_1 = 3$

$m_2 = -3$

real roots

$y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$

$y = c_1 e^{3x} + c_2 e^{-3x}$

$y_1 = e^{3x}$

$y_2 = e^{-3x}$

now identify linear independence by using wronskian

$w(y_1, y_2)$

$w = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$

$= y_1 y_2' - y_2 y_1'$

*calculate*

$y_1 = e^{3x}$

$y_1' = 3e^{3x}$

$y_2 = e^{-3x}$

$y_2' = -3e^{-3x}$

$w = \left(e^{3x}\right)\left(-3e^{-3x}\right) - \left(e^{-3x}\right)\left(3e^{3x}\right)$

$w = -3e^{0} - 3e^{0}$

$= -3 - 3$

$= -6$

$w = -6 \neq 0$

$w \neq 0$

$y_1$ and $y_2$ are linearly independent

$y = c_1 e^{3x} + c_2 e^{-3x}$

**Question # 4:** **Marks=05**

Find the second solution of the following differential equation by using the formula.

$x^2 y'' + 2xy' - 6y = 0, y_1 = x^2$

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**Answer:**

$y'' + P(x)y' + Q(x)y = 0$

*Given*

$x^2y'' + 2xy' - 6y = 0$

dividing by $x^2$ on both side

$y'' + \frac{2}{x}y' - \frac{6}{x^2}y = 0$

$p(x) = \frac{2}{3}$

$q(x) = -\frac{6}{x^2}$

*formula*

$y_2(x) = y_1(x)\int \frac{e^{-\int p(x)dx}}{(y_1(x))^2}dx$

now calculate

$\int p(x)dx$

$= \int \frac{2}{x}dx$

$= 2In|x|$

$= In(x^2)$

$e^{-\int p(x)dx} = e^{-In(x^2)} = \frac{1}{x^2}$

here we use formula

$y_2 = x^2\int \frac{\frac{1}{x^2}}{(x^2)^2}dx$

$y_2 = x^2\int \frac{1}{(x^2.x^4)}dx$

$y_2 = x^2\int \frac{1}{x^6}dx$

$y_2 = x^2\int x^{-6}dx$

$\int x^{-6}dx = \frac{x^{-5}}{-5}$

$= -\frac{1}{5}x^{-5}$

$y_2 = x^2\left(-\frac{1}{5}x^{-5}\right)$

$y_2 = -\frac{1}{5}x^{-3}$

$y_2 = x^{-3}$

$y = c_1x^2 + c_2x^{-3}$

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